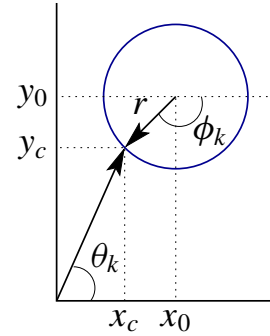




Illustrating the SIS steps with an example

- Probably, the SIS method still seems very abstract to you.
- Hopefully an example will help to illustrate it.¹
- Suppose there is a car driving around a circular track having radius 200 m.
- Velocity is nearly constant at 2° per sample, but may be perturbed by additive Gaussian noise modeling bumps, maneuvers, etc.
- We have a sensor located $500\sqrt{2}$ m from the center of the track that measures the angle to the car, θ_k (i.e., $z_k = \theta_k$) with noise $\pm 5^\circ$ typical.
- We wish to estimate ϕ_k (i.e., $x_k = \phi_k$).



¹Again, thanks to Prof. James McNames of Portland State University. I have adapted this example from his “State Space Tracking” course: <https://www.youtube.com/@JamesMcNames/videos>.



Process noise model

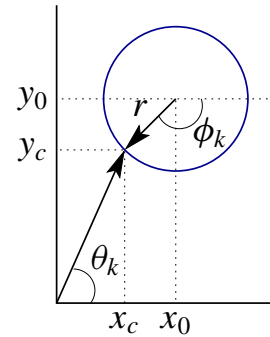
- For the process model, we write:

$$\phi_k = \phi_{k-1} + \omega + w_{k-1},$$

where $\omega = 2$ degrees/sample and $w_k \sim \mathcal{N}(0, \sigma_w^2)$.

- When implementing the SIS algorithm, we will need to know $f(x_k^{(i)} | x_{k-1}^{(i)})$, which is derived from this process model.
 - If $x_{k-1} = \phi_{k-1}$ is given, then $\phi_{k-1} + \omega$ is conditionally constant.
 - So, $x_k = \phi_k$ is equal to this constant plus w_k .
- Overall, the process model allows us to state:

$$f(x_k^{(i)} | x_{k-1}^{(i)}) = \mathcal{N}(x_{k-1}^{(i)} + \omega, \sigma_w^2).$$



Measurement noise model

- We model our measurement $z_k = \theta_k$ as:

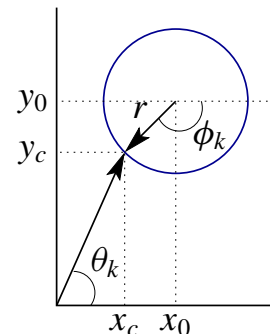
$$\theta_k = \tan^{-1} \left(\frac{y_{c,k}}{x_{c,k}} \right) + v_k,$$

where sensor noise $v_k \sim \mathcal{N}(0, \sigma_v^2)$.

- We will later need $f(z_k | x_k^{(i)})$, which is obtained from this measurement model; if $x_k = \phi_k$ is given, the $\tan^{-1}(\cdot)$ term is conditionally constant via the intermediate calculations:

$$x_{c,k} = r \cos(\phi_k) + x_0 \quad \text{and} \quad y_{c,k} = r \sin(\phi_k) + y_0.$$
- So, $z_k = \theta_k$ is equal to a constant plus the Gaussian RV v_k .
- Overall, the measurement model allows us to state:

$$f(z_k | x_k^{(i)}) = \mathcal{N} \left(\tan^{-1} \left(\frac{r \sin(x_k^{(i)}) + y_0}{r \cos(x_k^{(i)}) + x_0} \right), \sigma_v^2 \right).$$





Some final simulation constants

- Let $y_0 = x_0 = 500$ m, $r = 200$ m.
- The sensor noise is typically $\pm 5^\circ$ so we set $\sigma_v = 5^\circ$.
- We will set the process noise to have $\sigma_w = 10^\circ$.
- The mean initial state will be 50° and its noise will have standard deviation 10° .
- We conclude:

$$w_k \sim \mathcal{N}\left(0, \left(\frac{10\pi}{180}\right)^2\right) \quad \phi_0 \sim \mathcal{N}\left(\left(\frac{50\pi}{180}\right), \left(\frac{10\pi}{180}\right)^2\right).$$

$$v_k \sim \mathcal{N}\left(0, \left(\frac{5\pi}{180}\right)^2\right)$$



Design choices

- We have some design choices before we can employ the filter.
 - What importance density $q(x_k | x_{k-1}, z_k)$ to use?
 - Again using the substitutions $x_k = \phi_k$ and $z_k = \theta_k$, the easiest choice is $q(\phi_k | \phi_{k-1}, \theta_k) = f(\phi_k | \phi_{k-1}) \sim \mathcal{N}(\phi_{k-1} + 2, \sigma_w^2)$.
 - This choice results in a mathematical cancelation in the weight-update equation of $f(x_k | x_{k-1})/q(x_k | x_{k-1}, z_k)$ that simplifies the SIS implementation.
 - Number of particles:
 - Performance versus computation tradeoff.
 - Useful approach—as many as you have patience for.
 - When we implement SIS, we will use $N_p = 500$.
 - Estimator of state:
 - We'll use the mean, as it is easiest.



Summary

- The problem is now set up and ready to be implemented.
- The trickiest parts were determining the needed pdfs:
 - We find $f(x_k^{(i)} | x_{k-1}^{(i)})$ from the process model;
 - We find $f(z_k | x_k^{(i)})$ from the measurement model.
- My hope is that this example has helped to show how this can be done.
- We also had to design an importance density.
 - We chose $q(x_k^{(i)} | x_{k-1}^{(i)}, z_k) = f(x_k^{(i)} | x_{k-1}^{(i)})$.
 - Remember, this is a choice! We could have made other choices.
 - But, you will see in the next lesson that this is a very convenient choice to make when it is possible to do so.